A. Dataset Information

Fig. 3 shows randomly sampled images from each dataset, as well as the number of samples in each. Below we describe the methodology for constructing the Colonoscopy and Noise OOD datasets. We refer to the original papers of the remaining OOD datasets for their corresponding details [11, 13, 16, 17, 35]. We note, for ImageNet-200, there is a slight discrepancy between the number of samples reported in [17] and the number in the authors' provided datasets.[4] However, we do not believe this affects the validity of our results.

A.1. Noise

10,000 square images are generated to form this dataset. All random samples are generated independently. For each image, each pixel intensity value (across x,y,R,G,B) is sampled from the same distribution. It is a gaussian, $\mathcal{N}(0.5,\sigma^2)$ where σ differs *between* images. σ is generated by sampling from a unit gaussian and squaring the samples. Intensities are clipped to be in the range $[0,1]$ and mapped to integer values $\in \{0,...,255\}$. The image widths/heights are sampled from $\{2, ..., 256\}$ uniformly. Then, images are scaled to a resolution of 256×256 using the lanczos method.[5] The resulting images therefore vary in contrast, intensity and scale (see Fig. 3).

A.2. Colonoscopy

We separate out frames from the colonoscopical videos provided in [25].[6] The first 10 narrow band imaging (NBI) videos in each class of lesion (hyperplasic, serrated, adenoma) are downloaded. Each video frame is then extracted as an separate image to construct the dataset.

B. Model Training

To construct the Deep Ensemble, 5 independent training runs are performed, with random seeds $\{1, ..., 5\}$. ResNet-50 [8] is trained with the default hyperparameters in PyTorch's ImageNet example.[7] The ensemble members are trained on ImageNet-200 for 90 epochs with batch size equal to 256. Stochastic gradient descent is used with an initial learning rate of 0.1 that steps down by a factor of 10 at epochs 30 and 60. Weight decay is set to 10^{-4}, and momentum 0.9. During training images are augmented using `RandomHorizontalFlip` and `RandomResizedCrop`, and input to the network at a resolution of 224×224.

[4] https://github.com/daintlab/unknown-detection-benchmarks

[5] https://pillow.readthedocs.io/en/stable/_modules/PIL/Image.html#Image.resize

[6] http://www.depeca.uah.es/colonoscopy_dataset/

[7] https://github.com/pytorch/examples/tree/main/imagenet

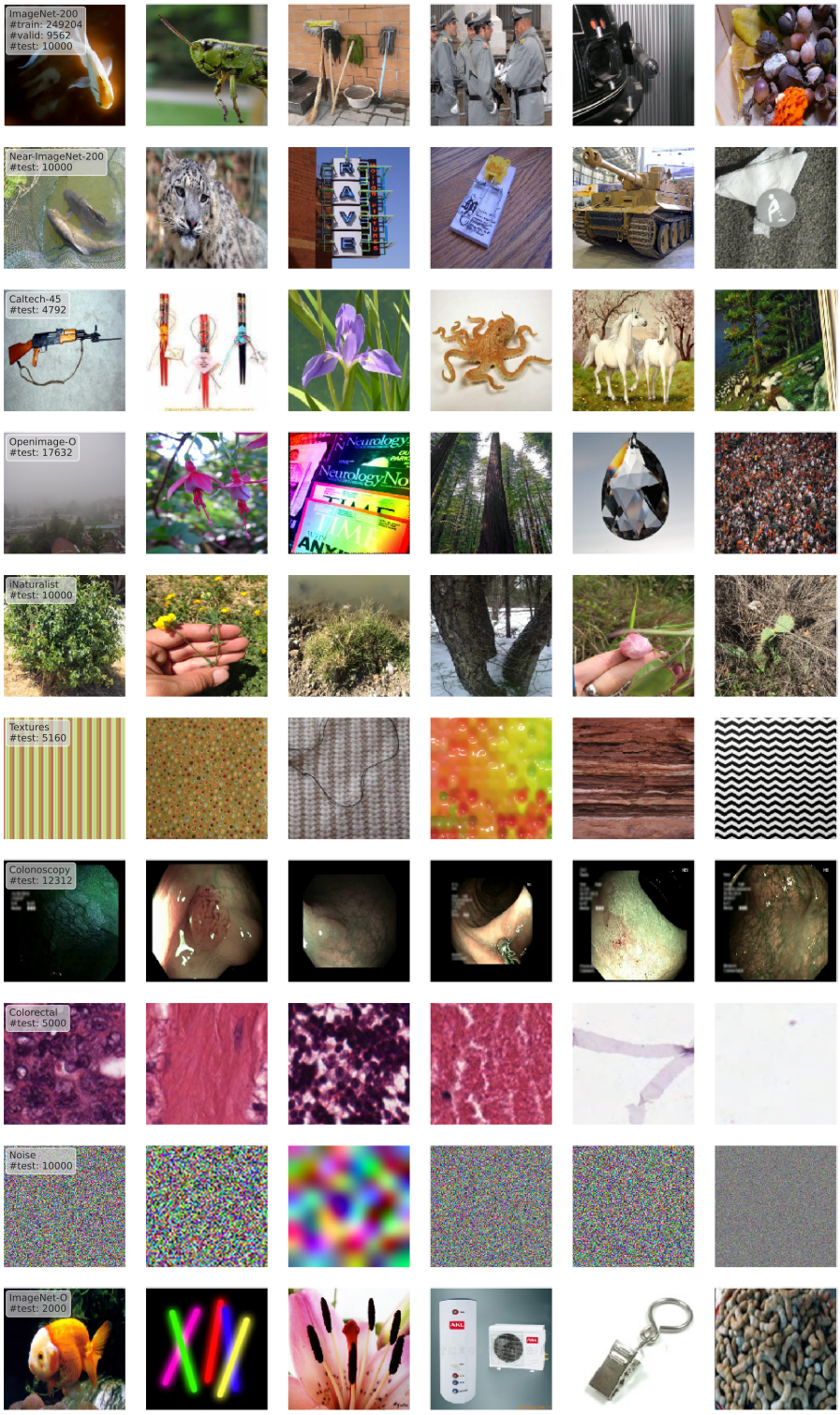

Figure 3. Example images from each dataset, with #samples in each.